

SipBit: A Sensing Platform to Recognize Beverage Type, Volume, and Sugar Content Using Electrical Impedance Spectroscopy and Deep Learning

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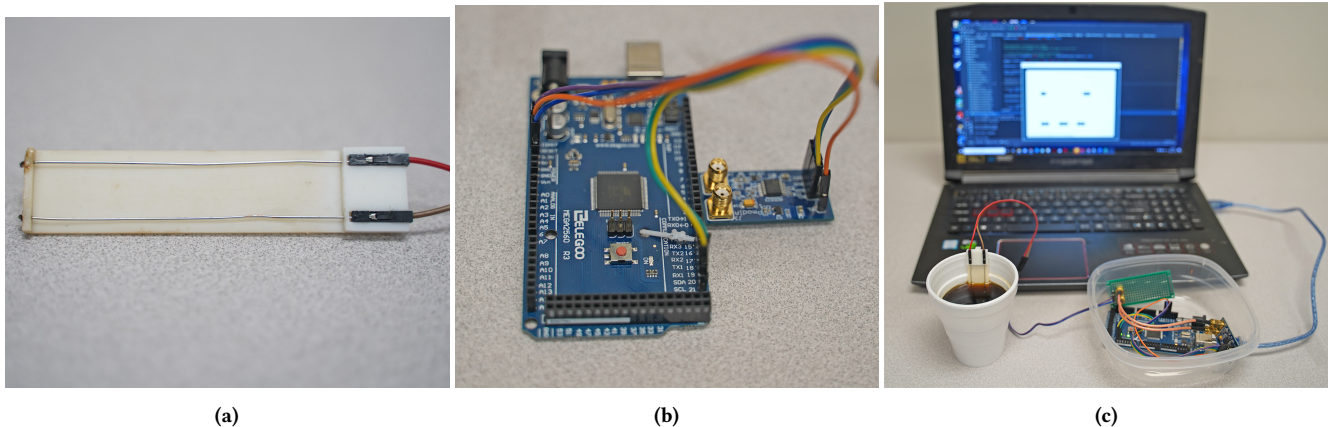


Figure 1: Components of SipBit: (a) 3D printed attachable module with electrodes, (b) impedance measuring module with control module, (c) and overall setup

ABSTRACT

We present SipBit, a sensing platform that digitally recognizes beverages and their attributes, an essential component in facilitating novel human-food interactions. SipBit consists of an electrical impedance measurement unit and a recognition method based on deep learning techniques. First, impedance measurements of a beverage are acquired using Electrical Impedance Spectroscopy. Then, a multi-task network cascade algorithm was employed to identify eight different beverage types in various volume levels and sugar concentrations. Results show that the multi-task network cascade discriminates beverage types with an accuracy of 96.32%, and estimates volumes with a root mean square error of 13.74ml and sugar content with a root mean square error of 7.99gdm^{-3} . Future work will include: 1) developing utensils embedded with SipBit for automatic beverage and attribute recognition, and 2) further developing SipBit to recognize additional beverage types and their attributes,

thus enabling a new avenue for designing human-food interactive technologies.

CCS CONCEPTS

• **Human-centered computing** → **Systems and tools for interaction design; Interaction devices;** • **Computing methodologies** → *Machine learning*.

KEYWORDS

Human-Food Interaction, Electrical Impedance Spectroscopy, Deep Learning, Multi-Task Learning

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1 INTRODUCTION

Interacting with food and beverages is an essential aspect of human life. With recent advances in digital technology, there has been increased incorporation of technology with food-related practices ranging from cultivation, cooking, presenting (serving), consumption, and food disposal. Integration of new digital technologies with

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food, including designing new interactions, are often discussed under the term Human-Food Interaction (HFI), where the interplay between technology, people, and food is explored [1]. When designing new HFI technologies, digital beverage sensing along with ingredient detection are integral components, yet it remains a significant challenge.

Popular sensing methods in contemporary literature employ computer vision-based techniques for digital food recognition [9, 10, 26]. These techniques typically recognize food items by capturing images of different meals and utilizing various image analysis methodologies, including deep learning, particularly Convolutional Neural Networks [13, 16, 32]. However, these computer vision-based techniques are limited when 1) quantifying the food since the depth information is harder to capture to estimate the volume of a food item, and 2) determining foods' internal attributes such as salt and sugar content since their variations are not reflected visually on the surface. Furthermore, it is even more challenging when employing these visual methods for beverage sensing as most beverages are visually similar and take the container's shape.

As a solution, we present SipBit - a beverage recognition and volume and ingredient (i.e., sugar content) estimation technology based on Electrical Impedance Spectroscopy (EIS) combined with deep learning techniques. EIS is a non-visual electrochemical measurement technique that has recently shown various applications in determining food quality and safety assessments [6]. However, despite the promising results in characterizing food and beverages, EIS and other non-visual techniques such as Electronic tongue (E-tongue [24]) and Electronic nose (E-nose [25]) are not widely explored in HFI research and consumer food technologies. Two primary reasons are sophisticated unwieldy device designs and variability of sensor measurements due to external factors such as temperature and moisture [20]. In addition, the integration of EIS and deep learning techniques to analyze sensor measurements have not been studied widely in the current literature.

Thus, we combine electrical impedance spectroscopy and deep learning to develop a novel beverage sensing system, as shown in Figure 1. Our approach is optimized to determine three main attributes of a beverage: 1) type of the beverage, 2) its volume, and 3) sugar content. A dataset was developed with 1221 instances (corresponding to the number of frequency sweeps) to evaluate our technology experimentally. It consists of impedance values, phase angles, real and imaginary parts of the impedances recorded from frequencies between 10kHz and 90kHz. Eight different beverage types (two types of black tea, two types of coffee, and four types of soda) were utilized to collect these instances along with different volumes and sugar content. Multi-task network cascade [3], a type of multi-task learning network, is employed in this study to determine three beverage attributes based on EIS data.

The main contributions of our work are as follows:

- Developing a non-visual approach for real-time beverage attribute recognition to facilitate the design of new human-food interactive technologies
- Combining EIS with deep multi-task network cascades to recognize beverage types and estimate volumes and sugar content

2 RELATED WORK

Contemporary research in Multimedia and Human-Computer Interaction (HCI) has integrated computer vision techniques to develop food sensing technologies, specifically focusing on applications such as dietary assessments [22, 29]. They primarily focused on recognizing food items in images or videos and estimating calorie content. For example, DietCam [10] recognizes food items using visual object recognition methods (combining a part-based model and texture verification model) to analyze the shape appearance, thereby classifying the data using Support Vector Machines (SVM). Furthermore, with recent advances in deep learning, researchers have employed deep learning techniques such as Convolution Neural Networks (CNN) in food identification [11, 13, 14, 16]. Unlike traditional feature extraction methods, CNN detects features in the image automatically without human supervision for subsequent regression or classification tasks [32].

There are multiple non-visual approaches primarily used for food analysis in food science though rarely employed in the context of HFI, including E-Tongue [25], E-Nose [21], and EIS-based [15] electrochemical measurement techniques. Among them, EIS is widely employed in recognizing attributes of food and beverages [4, 12, 23]. Although liquid sensing methodologies based on viscosity [7], radio frequency identification (RFID) [27, 28], and ultra-wideband signals [5] have been introduced in the mobile computing domain, the proposed methodologies have limitations in simultaneously determining liquid attributes. In addition, some of these approaches require larger instrumentation. Thus, unlike most automatic liquid sensing systems that focus on identifying only the beverage type, our research focuses on simultaneously discriminating beverages and recognizing multiple attributes: beverage type, quantity, and sugar content. In our work, Multi-task Network Cascade (MNC) [3] is selected as an adequate approach to simultaneously recognize beverage attributes, primarily due to the variability of the impedance spectrum.

3 METHODOLOGY

As shown in Figure 2, SipBit consists of four main components: 1) a 3D printed attachable module with electrodes, 2) an impedance measuring module (impedance analyzer), 3) the control module to program the impedance analyzer and communicate the measurements to a Personal Computer (PC), and 4) the PC to collect the measurements and process using a deep neural network to recognize the beverage attributes.

3.1 3D printed attachable module with electrodes

Figure 1a displays the 3D printed attachable electrode module with two aluminum electrodes that are placed 1.35cm apart and each 10cm in length. The electrode module is easily attachable to a beverage container such that it is submerged in the liquid, as shown in Figure 1c. This modular and attachable setup enables the collection of impedance measurements efficiently from multiple samples and beverage containers.

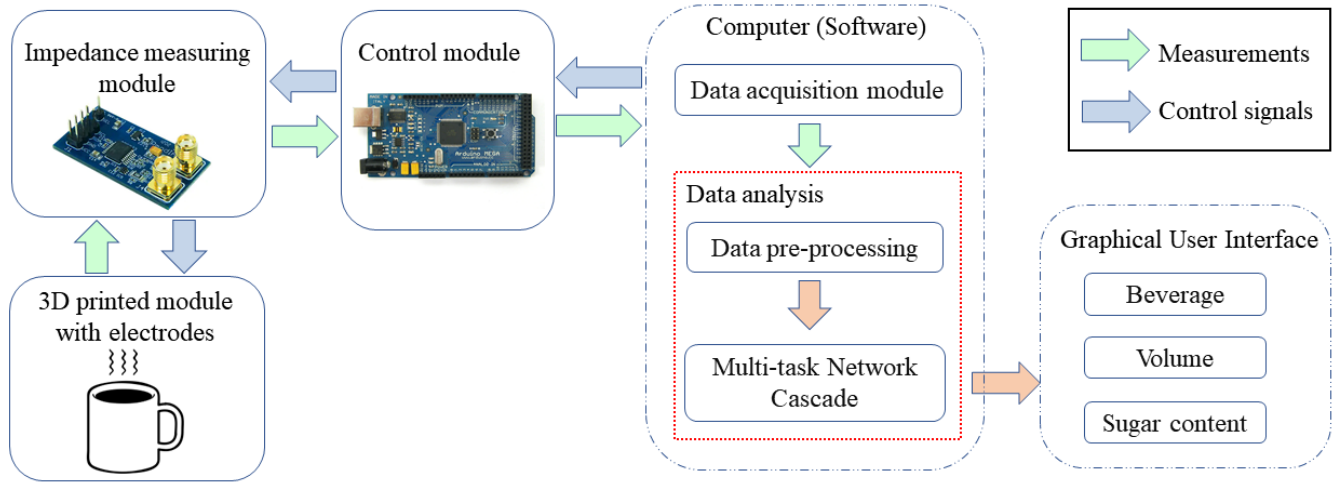


Figure 2: The system architecture of the SipBit system that utilizes EIS and deep multi-task learning networks for beverage and sugar content detection

3.2 Impedance measuring module

A Pmod impedance analyzer¹ with an AD5933 impedance converter² was employed to perform electrical impedance spectroscopy in the impedance measuring module. The impedance measuring module acquires impedance data of a beverage over 401 frequencies between 10kHz and 90kHz. The number of frequency points and the frequency range were selected based on the impedance analyzer's specifications. Once a frequency sweep is performed, it stores values that represent the real and imaginary parts of impedance in internal registers (real and imaginary registers).

3.3 Control module

The control module is configured using an Arduino Mega³ board that uses an Atmel ATmega2560 microcontroller. The control module is responsible for 1) configuring and activating the impedance analyzer to perform frequency sweeps, 2) extracting measured values from the impedance analyzer's real and imaginary registers, 3) calculating impedances, phase angles, real and imaginary parts of impedance based on values extracted from real and imaginary registers, and 4) transferring them to the PC. The Arduino Mega firmware was implemented based on the Arduino AD5933 library⁴ to handle the data transfer between the microcontroller and the impedance analyzer via I²C communication.

3.4 Software: Data acquisition, Analysis, Deep Learning Framework, and UI

The PC software is developed using Python programming language and consists of three sub-modules for data acquisition, analysis (including the Deep Learning Framework), and the User Interface (UI).

3.4.1 Data acquisition. The primary responsibilities of the data acquisition sub-module are to communicate with the control module (i.e., the Arduino Mega), calibrate the impedance analyzer, acquire impedance data via the serial interface (USART), and save the received data in a Comma-Separated Values (CSV) file.

3.4.2 Data analysis and deep learning framework. The data analysis process has three steps: 1) data pre-processing, 2) analyzing multi-task network cascades and selecting the best performing model, and 3) real-time testing.

Data pre-processing. The dataset has four types of features (i.e., impedance, phase angle, real and imaginary part of the impedance) that can be used to create 15 different combinations. First, we selected the feature combination of impedance and imaginary part of the impedance since this combination had shown the best results during our preliminary tests. Then, Principal Component Analysis (PCA) is applied on the selected feature combination. The principal component analysis is a dimension reduction technique that increases the data interpretability while minimizing the information loss [8]. Thus, at the end of the data pre-processing step, we selected two different feature combinations to evaluate the performance of two different multi-task network cascades (MNC).

Deep learning models (Multi-task network cascade). This study employed two types of multi-task network cascades (MNC) based on different features (results from the data pre-processing) to evaluate their performances. The feature combinations are 1) impedance and imaginary part of the impedance without applying PCA (MNC-Z-I) and 2) principal components of PCA analysis (MNC-PCA-Z-I) as shown in Figure 3. Each network consists of three blocks of layers, also known as stages, to identify the drink, determine the volume, and estimate the sugar content. Also, each stage has a loss term and is connected to a shared input layer. Loss terms depend on the outputs of the previous stages except for the loss term of the first stage. Therefore, the entire network was trained with a unified loss

¹<https://digilent.com/reference/pmod/pmodia/reference-manual>

²<https://www.analog.com/en/products/ad5933.html>

³<https://www.arduino.cc/en/Main/ArduinoBoardMega2560>

⁴<https://github.com/mjmeli/arduino-ad5933>

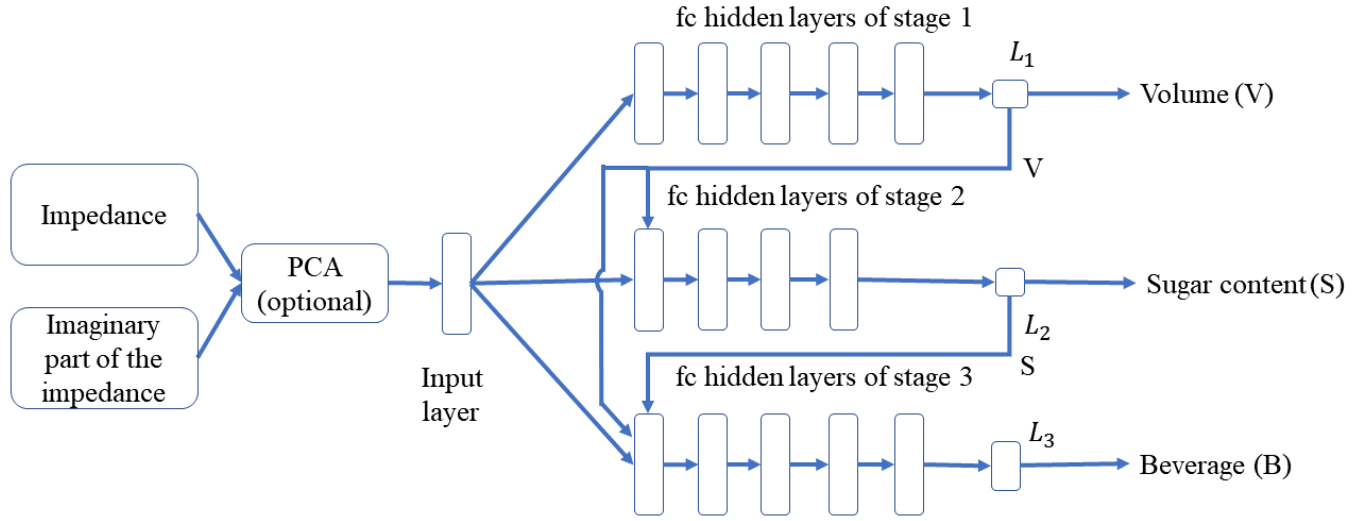


Figure 3: Multi-task network cascade used in the study. Two models can be developed based on the availability of PCA: 1) the model without PCA where impedances and imaginary parts of the impedances directly connected to the input layer, and 2) the model with PCA where impedances and imaginary parts of the impedance are processed by PCA and then connected to the input layer.

function as illustrated in equation 1.

$$L(\theta) = L_1(V(\theta)) + L_2(S(\theta)|V(\theta)) + L_3(B(\theta)|S(\theta), V(\theta)) \quad (1)$$

where $L(\theta)$ is the unified loss function, L_1 is loss in stage 1 (volume), L_2 is loss in stage 2 (sugar content), L_3 is loss in stage 3 (beverage), $V(\theta)$ is the output of stage 1 (volume predicted by the network), $S(\theta)$ is the output of stage 2 (sugar concentration predicted by the network), $B(\theta)$ is the output of stage 3 (beverage predicted by the model), and θ is the parameter to be optimized. Loss terms of the following stages and the unified loss term depend on outputs from previous stages. For instance, the gradient of the second loss function (L_3) involves gradient with regards to $V(\theta)$ and $S(\theta)$. All loss function equations were adopted from the study related to the multi-task network cascade [3]. In this study, we utilized sklearn⁵ and tensorflow⁶ libraries to implement the multi-task network cascades since they enable easy and fast development of neural networks.

Once trained, the network with the best performance was saved in Keras H5 format⁷ - a file format to save network parameters such as weights and node configurations.

3.4.3 Graphical User Interface (GUI). A Graphical User Interface (GUI) was developed to determine beverage attributes using the saved neural network in real-time. When the application starts, the saved deep learning model will be loaded. When a new beverage is available, new impedance values will be analyzed to recognize its attributes.

4 SYSTEM EVALUATION

To evaluate the effectiveness of this approach, we developed and optimized the deep learning framework for a total of eight beverages (four different beverage types and their variations) to identify their beverage type, volume, and sugar contents. Following are the beverages used in the analysis: 1) two types of black tea (LiptonTM and TwiningsTM English-Breakfast), 2) two types of coffee (StarbucksTM dark roasted and medium roasted), and 3) four types of soda (regular and diet coca-cola, and regular and diet Pepsi).

The impedance measurements were collected from a beverage (in two conditions: with and without sugar) in the following sequence. First, a beverage without sugar was poured at 30ml steps, and the impedance analyzer obtained measurements at each step. Once the volume reached 210ml, the liquid was poured into a beaker, and 5g of granulated sugar was dissolved in the beverage. Then, the beverage was poured back into the paper cup at 30ml steps, and the process of acquiring impedance measurements was continued until the volume reached 210ml. Next, 5g of granulated sugar was added after pouring the liquid into the beaker, and the process of impedance acquisition was repeated. This procedure was continued while adding sugar at 5g steps until a maximum of 25g. Sugar was not added to carbonated drinks; instead, the sugar content mentioned on the beverage's label was recorded.

The impedance acquisition process created a dataset that consists of 1221 records that correspond to the number of frequency sweeps. Furthermore, the dataset has 802 features corresponding to impedances and imaginary parts of impedances, each with values from 401 frequencies. It was split into training and test sets at a ratio of 80/20 (training: 977, test: 244). We evaluated multi-task

⁵<https://scikit-learn.org/stable/>

⁶<https://www.tensorflow.org/>

⁷https://www.tensorflow.org/guide/keras/save_and_serialize

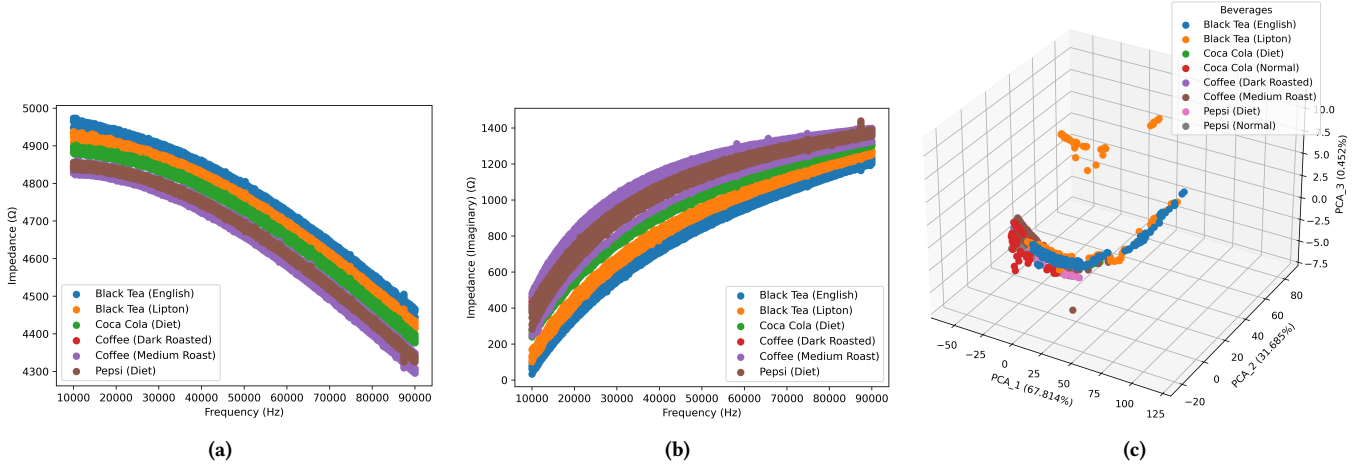


Figure 4: Results in preliminary measurements: (a) variation of impedance and (b) imaginary part of the impedance against frequency values are plotted for six types of beverages with a fixed volume of 30ml and sugar content of 0.00gdm^{-3} (no sugar), and (c) results of PCA analysis. Note that in a) and b) measurements of regular coca cola and pepsi are not included since they consist of sugar according to their labels and two diagrams represent beverages without sugar.

networks based on the performance of tasks (i.e., outputs). Regression tasks (estimating volume and sugar content) were evaluated using Root Mean Squared Error (RMSE), while the classification task (recognizing the beverage type) was evaluated using accuracy.

4.1 Preliminary measurements

The variations of impedances and imaginary part of the impedances acquired against the set of frequencies for four types of beverages (coffee - dark, medium, and black tea - Lipton, English) with a fixed volume of 30ml and sugar content of 0.00gdm^{-3} (no sugar) are illustrated in Figure 4a and Figure 4b. In addition, a similar analysis was conducted, and plots were generated for all other combinations of volumes and sugar content considering all the beverage types. Based on this analysis, we observed that when the frequency increases, the impedance variation shows a decreasing trend while the imaginary part of the impedance increases. However, it was challenging to discriminate between liquids as illustrated in Figure 4c. For example, Figure 4c demonstrates the results based on the first three principal components of PCA analysis that does not demonstrate a clear separation between beverages.

4.2 Multi-task network cascade

Table 1 shows the performance of MNC-Z-I and MNC-PCA-Z-I models while Figure 5 illustrates confusion matrices when discriminating beverages for the test dataset. The MNC-Z-I model performed beverage discrimination with training and test accuracies of 96.82% and 96.32%, respectively. In addition, it performs volume estimation with an RMSE of 13.74ml and 10.44ml for training and test datasets. Finally, the model performs sugar content assessment with an RMSE of 7.99gdm^{-3} and 5.74gdm^{-3} for test and training datasets. It is also important to note that the MNC-Z-I model demonstrates relatively lower over fittings in volume estimation.

The MNC-PCA-Z-I model exhibits improved performance in the classification task where accuracy was improved to 100% from 96.32% for the test dataset. Furthermore, it was observed that RMSE of volume and sugar content estimation has decreased relative to the MNC-Z-I model. However, volume estimation demonstrates overfitting where the RMSE for the test dataset is 90.10% higher than the training dataset. Therefore, the MNC-PCA-Z-I model is less effective in identifying beverage attributes compared to MNC-Z-I. Hence, based on our work, we conclude that the MNC-Z-I is the most effective network to discriminate the beverage type and estimate the volume and sugar concentration. However, as Table 1 and Figure 5 illustrates, both models MNC-Z-I and MNC-PCA-Z-I, warrant further evaluations with additional beverage types.

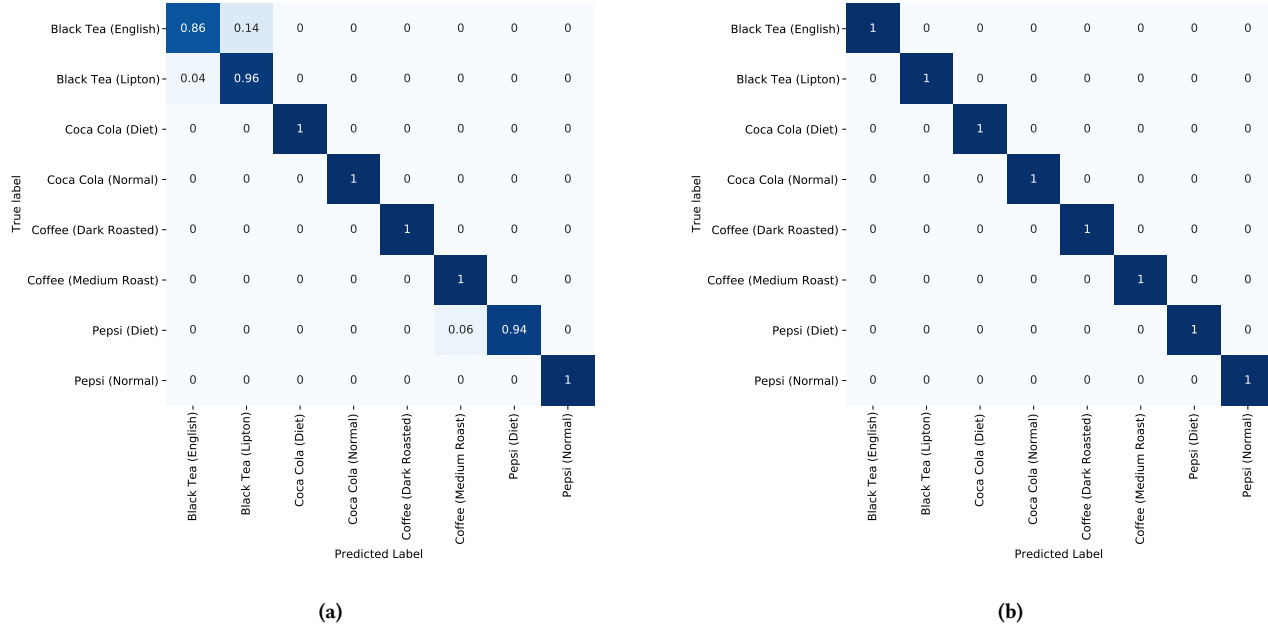
5 DISCUSSION

In comparison to existing technologies, our approach focuses on recognizing multiple attributes of a beverage. Combining EIS and multi-task network cascade, our approach focuses on simultaneously identifying beverage, volume, and sugar content, providing a platform technology to extend the recognition into other attributes of food and beverages and create novel human-food interfaces. Furthermore, since EIS measures the impedances between two electrodes, the container's shape that holds the liquid has less influence on the measurement. Therefore, EIS can recognize the beverage type, sugar content, and liquid level regardless of the container's shape.

Beverage type recognition shows promising results where the classification of the MNC-Z-I model reaches 96.32%. Furthermore, this network also shows promising results in two regression tasks with less overfitting, thus indicating that the MNC-Z-I model fits the training data and generalizes new data (test data) well. Our next step is to expand the dataset with more beverage types and their attributes and further evaluate the effectiveness of this approach.

Table 1: Performance of multi-task learning models (Z - impedance, and I - imaginary part of the impedance)

Model	Beverage classification		Volume estimation		Sugar concentration estimation	
	Training accuracy (%)	Testing accuracy (%)	RMSE (Training) (ml)	RMSE (Testing) (ml)	RMSE (Training) (gdm^{-3})	RMSE (Testing) (gdm^{-3})
MNC-Z-I	96.82	96.32	10.44	13.74	5.74	7.99
MNC-PCA-Z-I	100.00	100.00	5.86	11.14	4.89	5.34

**Figure 5: Normalized confusion matrices for a) MNC-Z-I and b) MNC-PCA-Z-I when discriminating beverages with the test dataset**

Although our technology takes a novel approach and shows promising results, improvements are required to address shortcomings such as overfitting in regression tasks. A potential reason for overfitting is the imbalance between instances and the number of features. Including measurements from more beverage samples in the dataset may improve the recognition performance of the multi-task network cascade to a great extent [30]. In addition, some solutions can be implemented from the neural network's perspective, such as regularization and dropout [30].

6 APPLICATIONS IN HUMAN-FOOD INTERACTIONS

We discuss several potential applications of SipBit in HFI under three themes: 1) food and beverage journaling, 2) virtual flavors, and 3) Wearable devices for non-invasive quality assessment.

6.1 Food and beverage journaling

Recording food and beverage intake could assist patients with chronic health conditions such as diabetes, hypertension, and kidney

diseases to manage their health by becoming aware of their food and beverage intake [2, 31]. Many people utilize digital food journaling technologies such as smartphone apps and websites. However, keeping a food journal (digital or on paper) is challenging and prone to errors (particularly for older adults who are often on restricted diets). In addition, when recording food data, users are unable to 1) quantify them accurately and 2) identify their internal composition (e.g., salt and sugar content). EIS-based food and ingredient detection approach could address these issues. For example, commonly used cutlery can be embedded with SipBit to automatically journal the users' food intake.

We envision several novel interactions based on the food and beverage journaling concept. For example, based on SipBit's real-time food intake monitoring, users may get real-time feedback related to their diets. Additionally, SipBit can also be integrated with activity trackers. These activity trackers can inform the activity levels the users are in each day, and then SipBit may recommend certain beverages and remind drinking water.

6.2 Virtual flavors

Recently, multiple research groups have attempted to simulate flavors via digital simulations. For example, in [19], authors combined electrical stimulation of tongue and smell modules that disperse aromas to simulate the multisensory flavor experiences while drinking plain water. Similar approaches have been proposed to simulate the flavor experience of various beverages like coffee [18] and lemonade [17]. One of the challenges in simulating virtual flavors is the inability to adjust digital sensory stimuli according to the beverage and its composition (e.g., ingredients). Now, the approach of SipBit can be used to create flavor profiles for different drinks, mapping beverages' electrical impedance to detect characteristics of food and beverages.

6.3 Wearable devices for non-invasive quality assessment

It is challenging for consumers to determine the quality of fruits and vegetables (e.g., ripeness and freshness) by visual inspection. Studies [6] have demonstrated that EIS is an effective and non-destructive technology in assessing the quality of food, particularly in laboratory environments. A methodology similar to SipBit could be developed as a wearable device for consumers to use during grocery shopping to determine fruits and vegetables' quality and other attributes effortlessly.

Another potential application area is food preparation. People rely on visual inspection and temperature when checking the status of food items during the cooking process. An alternative methodology based on SipBit technology could be developed to assess food items' cooking status in real-time, especially where consumers need food items to be cooked up to a certain status, such as the doneness of a steak.

7 CONCLUSION

We have presented SipBit, a novel digital approach to recognize beverages and estimate their volume levels and sugar content by combining electrical impedance spectroscopy and deep learning networks. In our work, although PCA improved the classification performance, the multi-task network cascade (MNC) without PCA-processed input features demonstrated better overall performance with lower overfitting. Immediate future work could expand the number of outputs to include additional ingredients such as salt, alcohol concentration, and total calorie content. Furthermore, integrating extra sensors such as thermal and chemical and combining additional types of analytical techniques such as infrared spectroscopy will improve the recognition performance. Finally, we discussed several application areas of SipBit, focusing Human-Food Interface developments.

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